

IFZAL HUSSAIN

COMPUTATIONAL MATERIALS SCIENTIST

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PROFILE

Researcher at Institut Jean Lamour, Universite de Lorraine, working on solidification dynamics through numerical simulation, mathematical modelling, and high-performance computing. Chemical engineer with experience spanning discrete element modelling, multiphase simulation, experimental research, and university teaching. Interested in connecting rigorous computational methods with observable material microstructures.

RESEARCH FOCUS

Solidification dynamics	Equiaxed grain growth	Numerical simulation
Mathematical modelling	High-performance computing	DEM
CFD	OpenFOAM	LIGGGHTS
COMSOL	Python	C++

PROFESSIONAL EXPERIENCE

Researcher

Oct 2024 - Present

Institut Jean Lamour (IJL), Universite de Lorraine | Nancy, France

- Develop numerical models of solidification dynamics and equiaxed dendritic grain growth.
- Use high-performance computing and mathematical modelling to improve model accuracy and connect simulations with material behaviour.
- Presented 'Simulation of Equiaxed Grain Growth in a Thin Sample' at MCWASP XVII in Winchester, United Kingdom (2026).

Marketing Associate

Jun 2024 - Sep 2024

Kuku FM | Bengaluru, India

- Launched audio shows and digital campaigns across Meta Ads and Google Ads.
- Led creative writers and experimented with AI-assisted promotional video workflows.

Teaching Assistant - Fluid Flow Lab

Jan 2024 - May 2024

Indian Institute of Technology Kharagpur | Kharagpur, India

- Taught second-year undergraduate students viscometer operation and fluid viscosity measurement; evaluated laboratory work and reports.

Teaching Assistant - Mechanical Operations Lab

Aug 2023 - Nov 2023

Indian Institute of Technology Kharagpur | Kharagpur, India

- Guided approximately 50 third-year undergraduates through ball mill fundamentals, operation, experimentation, and reporting.

Government Official and Physics Teacher

Feb 2018 - May 2022

Government of Jammu and Kashmir | Baramulla, India

- Taught physics to students in grades 9, 11, and 12 in an under-resourced community, maintaining a 100 percent examination success rate across four years.
- Helped increase the number of students choosing science from 3 in 2018 to 31 in 2022, while supporting school operations through the COVID-19 period.

Project Trainee

Oct 2016 - Mar 2017

Sewage Treatment Plants Srinagar | Srinagar, India

- Assessed plant operations, identified bottlenecks, and proposed strategies to reduce organic pollution and improve treatment performance for Dal Lake protection.

EDUCATION

Master of Technology - Chemical Engineering

2022 - 2024

Indian Institute of Technology Kharagpur | India

- Thesis: Representation of Non-spherical Particles in Discrete Element Method Simulations. Supervisor: Prof. Jayanta Chakraborty.
- Modelled super-quadric and multi-sphere particles in LIGGGHTS and investigated coarse-graining strategies to reduce computational cost.

Bachelor of Technology - Chemical Engineering

2012 - 2016

National Institute of Technology Srinagar | India

- Thesis: Enhancement of Heat Transfer in a Helical Coil Heat Exchanger Using CuO Nanofluid. Supervisor: Dr. Malik Parveez.
- Designed and fabricated the experimental heat exchanger and evaluated nanofluid heat-transfer performance.

SELECTED RESEARCH AND PROJECTS

Simulation of Equiaxed Grain Growth in a Thin Sample

MCWASP XVII conference presentation, Winchester, 2026.

Non-spherical Particles in DEM

Master's research using LIGGGHTS, super-quadrics, multi-sphere representations, and coarse-graining.

Microfluidic Droplet Formation

Two-phase COMSOL model of an unheated T-junction, examining flow-rate effects on droplet generation.

Dal Lake Sewage Treatment Optimisation

Six-month assessment of treatment performance and strategies to reduce effluent BOD.

Helical Coil Heat Exchanger

Experimental design and fabrication using CuO nanofluid to investigate enhanced heat transfer.

PUBLICATIONS AND WRITING

Nano Fluids - An Analysis. Book chapter in *Energy and Power*, 2021. ISBN 978-93-90781-04-1.

Author of the novellas *Imperfect Origami* (2018) and *Train on a Bridge* (2017).

LEADERSHIP AND WIDER PRACTICE

- Founder and Captain, NIT Srinagar Dramatic Society (2012-2016); grew the group to more than 100 members and staged over 10 productions.
- Travel photographer, theatre actor, and motorcycle explorer, bringing visual storytelling and curiosity into scientific communication.
- Tools for analysis and visualisation: NumPy, Pandas, ParaView, Gnuplot, Inkscape, GIMP, and DaVinci Resolve.